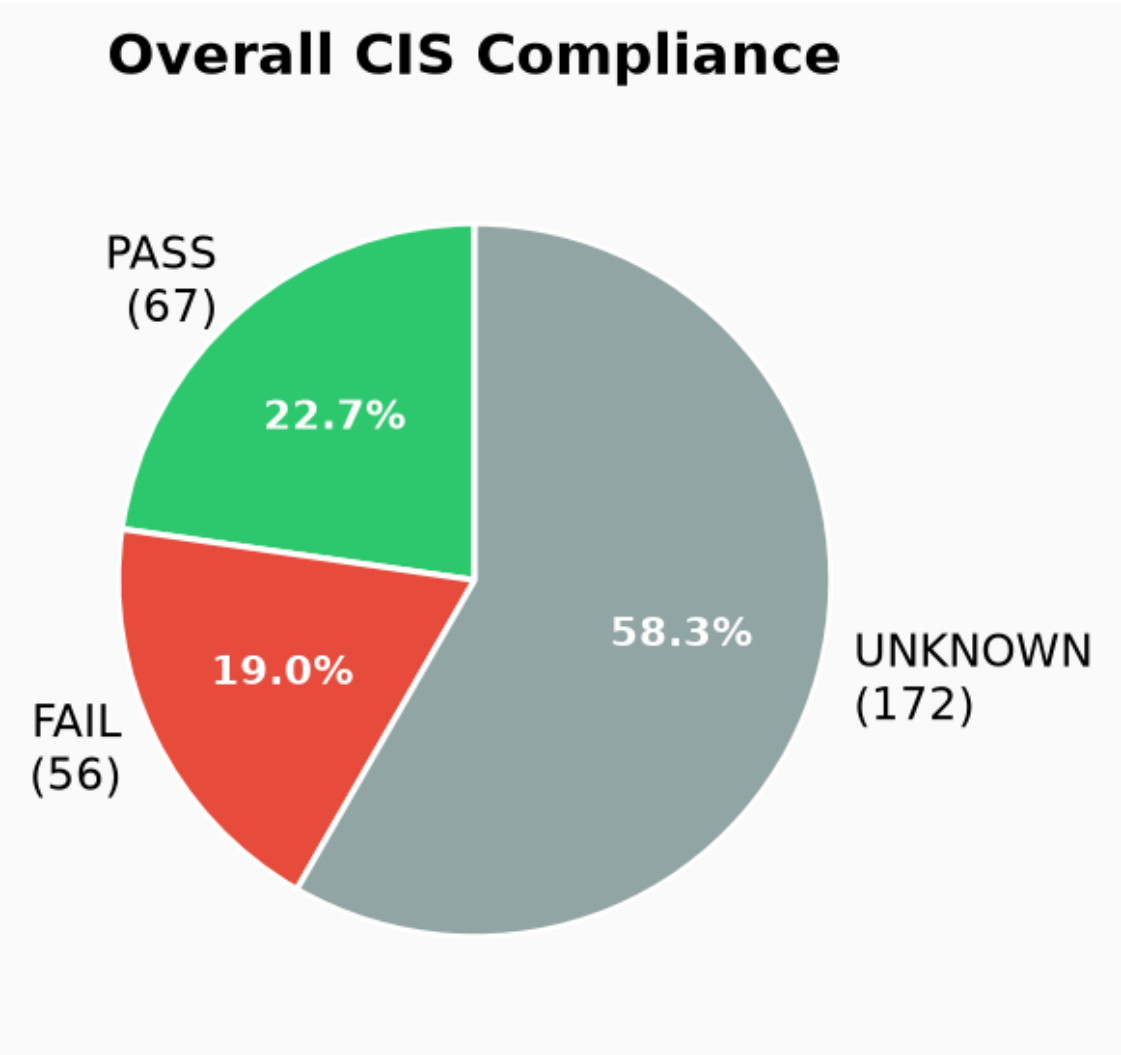


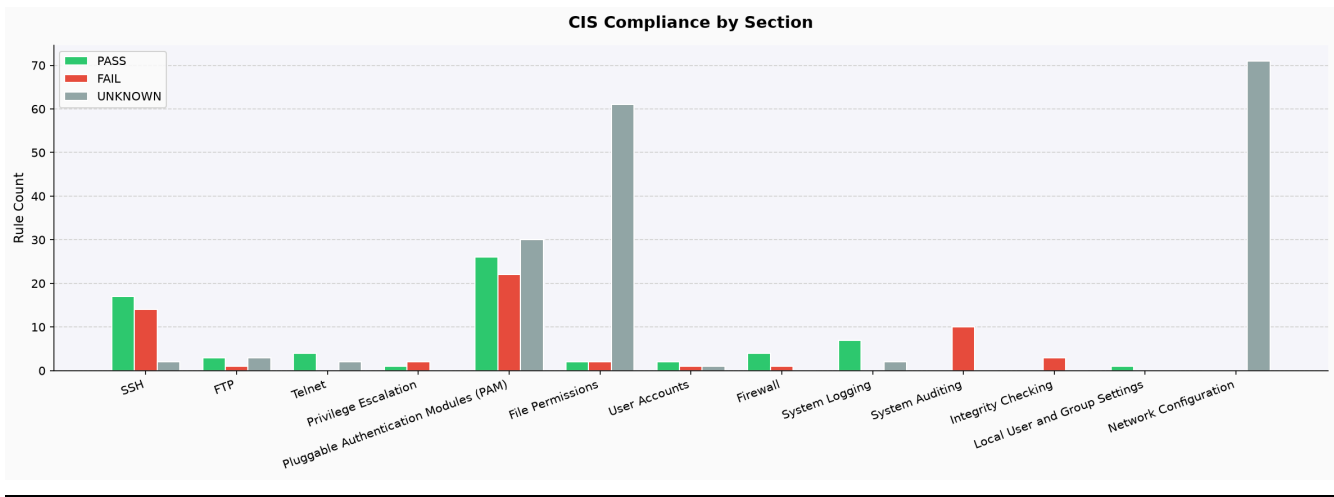
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	295
✓ Passed	67
✗ Failed	56
? Unknown	172
Overall Pass Rate	22.7%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
SSH	17	14	2	52%
FTP	3	1	3	43%
Telnet	4	0	2	67%
Privilege Escalation	1	2	0	33%
Pluggable Authentication Modules (PAM)	26	22	30	33%
File Permissions	2	2	61	3%
User Accounts	2	1	1	50%
Firewall	4	1	0	80%
System Logging	7	0	2	78%
System Auditing	0	10	0	0%
Integrity Checking	0	3	0	0%
Local User and Group Settings	1	0	0	100%
Network Configuration	0	0	71	0%

CIS Benchmark Compliance Report

This report details the compliance status of the remote machine against the specified CIS Ubuntu 24.04 LTS Benchmark rules.

Note on Data Collection Errors: A significant portion of the security profile data (including filesystem, package_management, network_config, apparmor, bootloader, process_hardening, warning_banners, gnome, services, time_sync, job_schedulers) reported json_parse errors. This indicates that the data for these sections could not be collected or processed correctly. Consequently, all rules relying on these data points are marked as **UNKNOWN**.

SSH Rules

Rule 1.6.5: Ensure sshd warning Banner is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is set to "none".
- **Recommendation:** Configure a warning banner in `/etc/ssh/sshd_config` by setting `Banner /etc/issue.net` or a custom banner file.

Rule 1.6.10: Ensure access to sshd warning banner is configured

- **Status:** INFORMATIONAL
- **Evidence:** `ssh.features.banner` is "none". No banner file is configured, so its access permissions cannot be checked.
- **Recommendation:** First, configure a warning banner (see Rule 1.6.5). Then, ensure the banner file has permissions `0644` or less, owned by `root:root`.

Rule 5.1.1: Ensure access to /etc/ssh/sshd_config is configured

- **Status:** PASS
- **Evidence:** `/etc/ssh/sshd_config` has permissions `644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 5.1.2: Ensure access to SSH private host key files is configured

- **Status:** PASS
- **Evidence:** All SSH private host key files (`/etc/ssh/ssh_host_ecdsa_key`, `/etc/ssh/ssh_host_ed25519_key`, `/etc/ssh/ssh_host_rsa_key`) have permissions `600`, owner `root`, group `root`.
- **Recommendation:** None

Rule 5.1.3: Ensure access to SSH public host key files is configured

- **Status:** PASS
- **Evidence:** All SSH public host key files (`/etc/ssh/ssh_host_ecdsa_key.pub`, `/etc/ssh/ssh_host_ed25519_key.pub`, `/etc/ssh/ssh_host_rsa_key.pub`) have permissions `644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 5.1.4: Ensure sshd access is configured

- **Status:** INFORMATIONAL
- **Evidence:** This is a general rule covered by more specific SSH configuration checks.
- **Recommendation:** None

Rule 5.1.5: Ensure sshd Banner is configured

- **Status:** FAIL
- **Evidence:** `ssh.features.banner` is set to "none".
- **Recommendation:** Configure a warning banner in `/etc/ssh/sshd_config` by setting `Banner /etc/issue.net` or a custom banner file.

Rule 5.1.6: Ensure sshd Ciphers are configured

- **Status:** PASS
- **Evidence:** Configured ciphers are `chacha20-poly1305@openssh.com`, `aes128-ctr`, `aes192-ctr`, `aes256-ctr`, `aes128-gcm@openssh.com`, `aes256-gcm@openssh.com`. These are considered strong ciphers.
- **Recommendation:** None

Rule 5.1.7: Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured

- **Status:** FAIL
- **Evidence:** `ClientAliveInterval` is 0. `ClientAliveCountMax` is 3.
- **Recommendation:** Set `ClientAliveInterval` to a reasonable non-zero value (e.g., 300) in `/etc/ssh/sshd_config` to terminate idle sessions.

Rule 5.1.8: Ensure sshd DisableForwarding is enabled

- **Status:** FAIL
- **Evidence:** `AllowTcpForwarding` is `true`, `AllowAgentForwarding` is `true`, and `X11Forwarding` is `true`.
- **Recommendation:** Set `AllowTcpForwarding no`, `AllowAgentForwarding no`, and `X11Forwarding no` in `/etc/ssh/sshd_config` to disable forwarding features.

Rule 5.1.9: Ensure sshd GSSAPIAuthentication is disabled

- **Status:** PASS
- **Evidence:** `GSSAPIAuthentication` is `false`.
- **Recommendation:** None

Rule 5.1.10: Ensure sshd HostbasedAuthentication is disabled

- **Status:** PASS
- **Evidence:** `HostbasedAuthentication` is `false`.
- **Recommendation:** None

Rule 5.1.11: Ensure sshd IgnoreRhosts is enabled

- **Status:** PASS
- **Evidence:** `IgnoreRhosts` is `true`.
- **Recommendation:** None

Rule 5.1.12: Ensure sshd KexAlgorithms is configured

- **Status:** FAIL
- **Evidence:** `crypto.weak_kex` lists `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`, and `diffie-hellman-group14-sha256` as weak algorithms. CVE-2002-20001 (DROWN) is also listed as a high risk vulnerability.
- **Recommendation:** Explicitly configure `KexAlgorithms` in `/etc/ssh/sshd_config` to exclude weak algorithms and prioritize strong, modern key exchange algorithms.

Rule 5.1.13: Ensure sshd LoginGraceTime is configured

- **Status:** FAIL
- **Evidence:** `LoginGraceTime` is 120.
- **Recommendation:** Set `LoginGraceTime` 60 or less in `/etc/ssh/sshd_config`.

Rule 5.1.14: Ensure sshd LogLevel is configured

- **Status:** PASS
- **Evidence:** `LogLevel` is INFO.
- **Recommendation:** None

Rule 5.1.15: Ensure sshd MACs are configured

- **Status:** FAIL
- **Evidence:** `crypto.weak_macs` lists multiple weak MAC algorithms, including `hmac-sha1-etm@openssh.com` which is affected by CVE-2023-48795 (Terrapin Attack).
- **Recommendation:** Explicitly configure MACs in `/etc/ssh/sshd_config` to exclude weak algorithms and prioritize strong, modern Message Authentication Codes.

Rule 5.1.16: Ensure sshd MaxAuthTries is configured

- **Status:** FAIL
- **Evidence:** `MaxAuthTries` is 6.
- **Recommendation:** Set `MaxAuthTries` 4 or less in `/etc/ssh/sshd_config`.

Rule 5.1.17: Ensure sshd MaxStartups is configured

- **Status:** FAIL
- **Evidence:** `MaxStartups` is 10:30:100.
- **Recommendation:** Set `MaxStartups` 10:30:60 or a similar value to limit concurrent unauthenticated connections in `/etc/ssh/sshd_config`.

Rule 5.1.18: Ensure sshd MaxSessions is configured

- **Status:** PASS
- **Evidence:** `MaxSessions` is 10.
- **Recommendation:** None

Rule 5.1.19: Ensure sshd PermitEmptyPasswords is disabled

- **Status:** PASS
- **Evidence:** `PermitEmptyPasswords` is false.
- **Recommendation:** None

Rule 5.1.20: Ensure sshd PermitRootLogin is disabled

- **Status:** FAIL
- **Evidence:** `PermitRootLogin` is `without-password`.
- **Recommendation:** Set `PermitRootLogin` `no` in `/etc/ssh/sshd_config` to completely disable direct root login.

Rule 5.1.21: Ensure sshd PermitUserEnvironment is disabled

- **Status:** PASS
- **Evidence:** `PermitUserEnvironment` is false.
- **Recommendation:** None

Rule 5.1.22: Ensure sshd UsePAM is enabled

- **Status:** PASS
- **Evidence:** `UsePAM` is true.
- **Recommendation:** None

Rule 5.1.23: Ensure sshd post-quantum cryptography key exchange algorithms are configured

- **Status:** FAIL
- **Evidence:** The `crypto.weak_kex` section indicates that several non-PQC KEX algorithms with known weaknesses are present. There is no evidence of PQC KEX algorithms being configured.
- **Recommendation:** Configure `KexAlgorithms` in `/etc/ssh/sshd_config` to include and prioritize post-quantum cryptography key exchange algorithms, and remove any weak or deprecated algorithms.

Rule 5.1.24: Ensure sshd ListenAddress is configured

- **Status:** FAIL
- **Evidence:** `ListenAddress` is `0.0.0.0:22`.
- **Recommendation:** Set `ListenAddress` to a specific IP address (e.g., `192.168.1.10:22`) in `/etc/ssh/sshd_config` to restrict SSH access to a specific interface.

FTP Rules

Rule 2.1.8: Ensure ftp server services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure no FTP server packages (e.g., `vsftpd`, `proftpd`, `pure-ftpd`) are installed and no FTP server units are active.

Rule 2.1.20: Ensure tftp server services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure no TFTP server packages (e.g., `tftpd-hpa`) are installed and no TFTP server units are active.

Rule 2.2.6: Ensure ftp client is not installed

- **Status:** UNKNOWN

- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure no FTP client packages (e.g., `ftp`) are installed.

TELNET Rules

Rule 2.1.19: Ensure telnet server services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure no Telnet server packages (e.g., `telnetd`) are installed and no Telnet server units are active.

Rule 2.2.4: Ensure telnet client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure no Telnet client packages (e.g., `telnet`) are installed.

ESCALATION Rules

Rule 5.2.1: Ensure sudo is installed

- **Status:** PASS
- **Evidence:** `sudo_installed` is `true`, version `1.9.15p5`.
- **Recommendation:** None

Rule 5.2.2: Ensure sudo commands use pty

- **Status:** PASS
- **Evidence:** `/etc/sudoers` contains `Defaults use_pty`.
- **Recommendation:** None

Rule 5.2.3: Ensure sudo log file exists

- **Status:** PASS
- **Evidence:** `auth.log` and `syslog` exist, and `rsyslog_authpriv_configured` is `true`. Journald also shows sudo evidence.
- **Recommendation:** None

Rule 5.2.4: Ensure users must provide password for escalation

- **Status:** PASS
- **Evidence:** `nopasswd_entries` is empty, indicating no NOPASSWD configurations.
- **Recommendation:** None

Rule 5.2.5: Ensure re-authentication for privilege escalation is not

disabled globally

- **Status:** PASS
- **Evidence:** `noauthenticate_entries` is empty, indicating no global `NOAUTHENTICATE` configurations.
- **Recommendation:** None

Rule 5.2.6: Ensure `sudo timestamp_timeout` is configured

- **Status:** FAIL
- **Evidence:** `timestamp_timeout_entries` is empty, meaning `timestamp_timeout` is not explicitly configured.
- **Recommendation:** Add `Defaults timestamp_timeout=0` (to require password every time) or `Defaults timestamp_timeout=15` (to allow re-use for 15 minutes) to `/etc/sudoers` or a file in `/etc/sudoers.d/`.

Rule 5.2.7: Ensure access to the `su` command is restricted

- **Status:** FAIL
- **Evidence:** The `pam_wheel.so` module is commented out in `/etc/pam.d/su`, meaning `su` access is not restricted to the `wheel` group.
- **Recommendation:** Uncomment `auth required pam_wheel.so` in `/etc/pam.d/su` and ensure only authorized users are members of the `wheel` group.

DIRECTORY ACCESS Rules

Rule 7.1.1: Ensure access to `/etc/passwd` is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd` has permissions `0644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.2: Ensure access to `/etc/passwd-` is configured

- **Status:** PASS
- **Evidence:** `/etc/passwd-` has permissions `0644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.3: Ensure access to `/etc/group` is configured

- **Status:** PASS
- **Evidence:** `/etc/group` has permissions `0644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.4: Ensure access to `/etc/group-` is configured

- **Status:** PASS
- **Evidence:** `/etc/group-` has permissions `0644`, owner `root`, group `root`.
- **Recommendation:** None

Rule 7.1.5: Ensure access to `/etc/shadow` is configured

- **Status:** PASS
- **Evidence:** /etc/shadow has permissions 0640, owner root, group shadow.
- **Recommendation:** None

Rule 7.1.6: Ensure access to /etc/shadow- is configured

- **Status:** PASS
- **Evidence:** /etc/shadow- has permissions 0640, owner root, group shadow.
- **Recommendation:** None

Rule 7.1.7: Ensure access to /etc/gshadow is configured

- **Status:** PASS
- **Evidence:** /etc/gshadow has permissions 0640, owner root, group shadow.
- **Recommendation:** None

Rule 7.1.8: Ensure access to /etc/gshadow- is configured

- **Status:** PASS
- **Evidence:** /etc/gshadow- has permissions 0640, owner root, group shadow.
- **Recommendation:** None

Rule 7.1.9: Ensure access to /etc/shells is configured

- **Status:** PASS
- **Evidence:** /etc/shells has permissions 0644, owner root, group root.
- **Recommendation:** None

Rule 7.1.10: Ensure access to /etc/security/opasswd is configured

- **Status:** PASS
- **Evidence:** /etc/security/opasswd has permissions 0600, owner root, group root.
- **Recommendation:** None

Rule 7.1.11: Ensure world writable files and directories are secured

- **Status:** PASS
- **Evidence:** No world-writable files or directories without the sticky bit were found on scanned mounts.
- **Recommendation:** None

Rule 7.1.12: Ensure no files or directories without an owner and a group exist

- **Status:** PASS
- **Evidence:** No unowned or ungrouped files or directories were found on scanned mounts.
- **Recommendation:** None

Rule 7.1.13: Ensure SUID and SGID files are reviewed

- **Status:** INFORMATIONAL
- **Evidence:** The following SUID/SGID files were found: `/usr/bin/umount`, `/usr/bin/chsh`, `/usr/bin/sudo`, `/usr/bin/su`, `/usr/bin/passwd`, `/usr/bin/ssh-agent`, `/usr/bin/fusermount3`, `/usr/bin/newgrp`, `/usr/bin/chage`, `/usr/bin/expiry`, `/usr/bin/gpasswd`, `/usr/bin/crontab`, `/usr/bin/mount`, `/usr/bin/chfn`, `/usr/lib/dbus-1.0/dbus-daemon-launch-helper`, `/usr/lib/x86_64-linux-gnu/utempter/utempter`, `/usr/lib/openssh/ssh-keysign`, `/usr/lib/polkit-1/polkit-agent-helper-1`, `/usr/sbin/pam_extrausers_chkpwd`, `/usr/sbin/unix_chkpwd`.
- **Recommendation:** Manually review the listed SUID/SGID files to ensure they are legitimate and necessary for system operation.

USER ACCOUNTS Rules

Rule 5.4.1.1: Ensure password expiration is configured

- **Status:** FAIL
- **Evidence:** `PASS_MAX_DAYS` is set to 99999 in `/etc/login.defs`, meaning passwords never expire.
- **Recommendation:** Set `PASS_MAX_DAYS` to 365 or less in `/etc/login.defs` to enforce password expiration.

Rule 5.4.1.2: Ensure minimum password days is configured

- **Status:** INFORMATIONAL
- **Evidence:** `PASS_MIN_DAYS` is set to 0 in `/etc/login.defs`, allowing immediate password changes.
- **Recommendation:** Manually review and consider setting `PASS_MIN_DAYS` to 1 or more in `/etc/login.defs` to prevent users from cycling through passwords.

Rule 5.4.1.3: Ensure password expiration warning days is configured

- **Status:** PASS
- **Evidence:** `PASS_WARN_AGE` is set to 7 in `/etc/login.defs`.
- **Recommendation:** None

Rule 5.4.1.4: Ensure strong password hashing algorithm is configured

- **Status:** PASS
- **Evidence:** `ENCRYPT_METHOD` is SHA512 in `/etc/login.defs`, and `yescrypt` is used by `pam_unix.so`.
- **Recommendation:** None

Rule 5.4.1.5: Ensure inactive password lock is configured

- **Status:** FAIL
- **Evidence:** `INACTIVE` is set to -1 in `/etc/default/useradd` (implied by `useradd_inactive`), meaning no inactive lock.
- **Recommendation:** Set `INACTIVE` to 30 or less in `/etc/default/useradd` to lock accounts after a period of inactivity.

Rule 5.4.1.6: Ensure all users last password change date is in the past

- **Status:** FAIL
- **Evidence:** Multiple users (e.g., root, daemon, bin, sys, sync, games, man, lp, mail, news, uucp, proxy, www-data, backup, list, irc, _apt, nobody, systemd-network, systemd-timesync, dhcpcd, messagebus, systemd-resolve, pollinate, polkitd, syslog, uidd, tcpdump, tss, landscape, fwupd-refresh, usbmux, sebt, wazuh-indexer, wazuh, wazuh-dashboard, sshd, ftp) have their last_password_change date set in the future.
- **Recommendation:** Force a password change for all affected users, or manually adjust their last_password_change dates to a past date using `chage -d YYYY-MM-DD <username>`.

Rule 5.4.2.1: Ensure root is the only UID 0 account

- **Status:** PASS
- **Evidence:** Only root is listed as a UID 0 account.
- **Recommendation:** None

Rule 5.4.2.2: Ensure root is the only GID 0 account

- **Status:** PASS
- **Evidence:** Only root is listed as a primary GID 0 account.
- **Recommendation:** None

Rule 5.4.2.3: Ensure group root is the only GID 0 group

- **Status:** PASS
- **Evidence:** Only root is listed as a GID 0 group.
- **Recommendation:** None

Rule 5.4.2.4: Ensure root account access is controlled

- **Status:** FAIL
- **Evidence:** PermitRootLogin is without-password (not no). There is no securetty file to restrict console access.
- **Recommendation:** Set PermitRootLogin no in /etc/ssh/sshd_config. Consider creating /etc/securetty to restrict root console logins to specific TTYS.

Rule 5.4.2.5: Ensure root path integrity

- **Status:** PASS
- **Evidence:** Root's PATH contains standard system directories. No world-writable directories were found in 7.1.11.
- **Recommendation:** None

Rule 5.4.2.6: Ensure root user umask is configured

- **Status:** PASS
- **Evidence:** UMASK is 022 in /etc/login.defs. No conflicting umask settings found in root's shell profiles.
- **Recommendation:** None

Rule 5.4.2.7: Ensure system accounts do not have a valid login shell

- **Status:** FAIL
- **Evidence:** The `sync` user has `/bin/sync` as its shell, which is a valid login shell.
- **Recommendation:** Change the shell for the `sync` user to `/usr/sbin/nologin` or `/bin/false` using `usermod -s /usr/sbin/nologin sync`.

Rule 5.4.2.8: Ensure accounts without a valid login shell are locked

- **Status:** PASS
- **Evidence:** All system accounts listed have a `passwd_status` of `L` (locked).
- **Recommendation:** None

Rule 5.4.3.1: Ensure nologin is not listed in /etc/shells

- **Status:** PASS
- **Evidence:** `/usr/sbin/nologin` or `/bin/false` are not listed in `/etc/shells`.
- **Recommendation:** None

Rule 5.4.3.2: Ensure default user shell timeout is configured

- **Status:** FAIL
- **Evidence:** No `TMOUT` setting was found in `/etc/profile`, `/etc/profile.d/`, or `/etc/bash.bashrc`.
- **Recommendation:** Add `TMOUT=600` (or a lower value) to `/etc/profile.d/timeout.sh` to automatically log out idle shell sessions.

Rule 5.4.3.3: Ensure default user umask is configured

- **Status:** PASS
- **Evidence:** `UMASK` is `022` in `/etc/login.defs`, and `pam_umask.so` is configured in `common-session`.
- **Recommendation:** None

FIREWALL Rules

Rule 4.1.1: Ensure ufw is installed

- **Status:** PASS
- **Evidence:** `ufw_installed` is `true`, `ufw_dpkg_status` is `"install ok installed"`.
- **Recommendation:** None

Rule 4.1.2: Ensure ufw service is configured

- **Status:** FAIL
- **Evidence:** `ufw` service is enabled and reported as active by `systemd`, but `ufw status` reports `"Status: inactive"`.
- **Recommendation:** Activate the UFW firewall using `sudo ufw enable`.

Rule 4.1.3: Ensure ufw incoming default is configured

- **Status:** PASS
- **Evidence:** `DEFAULT_INPUT_POLICY` is `DROP` in `/etc/default/ufw`.
- **Recommendation:** None

Rule 4.1.4: Ensure ufw outgoing default is configured

- **Status:** PASS
- **Evidence:** `DEFAULT_OUTPUT_POLICY` is `ACCEPT` in `/etc/default/ufw`.
- **Recommendation:** None (for Level 1). For Level 2, consider setting to `DROP` and explicitly allowing necessary outgoing connections.

Rule 4.1.5: Ensure ufw routed default is configured

- **Status:** PASS
- **Evidence:** `DEFAULT_FORWARD_POLICY` is `DROP` in `/etc/default/ufw`.
- **Recommendation:** None

SYSTEM LOGGING Rules

Rule 6.1.1.1.1: Ensure systemd-journald service is enabled and active

- **Status:** PASS
- **Evidence:** `systemd-journald` service is static and active.
- **Recommendation:** None

Rule 6.1.1.1.2: Ensure systemd-journal-remote service is not in use

- **Status:** PASS
- **Evidence:** `systemd-journal-remote` is not installed, `systemd-journal-upload` is inactive, and `systemd-journal-remote.socket` is inactive.
- **Recommendation:** None

Rule 6.1.1.1.3: Ensure journald is configured to send logs to rsyslog

- **Status:** PASS
- **Evidence:** `ForwardToSyslog` is commented out in `/etc/systemd/journald.conf`, implying the default behavior which is to forward to syslog if available. `rsyslog` is installed and active.
- **Recommendation:** None

Rule 6.1.1.1.4: Ensure journald log file access is configured

- **Status:** INFORMATIONAL
- **Evidence:** Journal directories `/run/log/journal` and `/var/log/journal` have permissions `2755`, owner `root`, group `systemd-journal` (GID `999`). Journal files have `0640`, owner `root`, group `systemd-journal`.
- **Recommendation:** Manually verify that these permissions are appropriate for the environment and that only authorized users/groups have read access to sensitive log data.

Rule 6.1.1.1.5: Ensure journald log file rotation is configured

- **Status:** INFORMATIONAL
- **Evidence:** Journald log rotation parameters (`SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, `SystemMaxFiles`, `MaxRetentionSec`, `MaxFileSec`) are commented out in `/etc/systemd/journald.conf`, relying on defaults.
- **Recommendation:** Manually review the default journald log rotation settings and configure them explicitly in `/etc/systemd/journald.conf` if the defaults do not meet organizational requirements for log retention and disk usage.

Rule 6.1.1.1.6: Ensure journald Storage is configured

- **Status:** PASS
- **Evidence:** Storage is commented out in `/etc/systemd/journald.conf`, defaulting to `auto`. Since `/var/log/journal` exists, storage is persistent.
- **Recommendation:** None

Rule 6.1.1.1.7: Ensure journald Compress is configured

- **Status:** PASS
- **Evidence:** Compress is commented out in `/etc/systemd/journald.conf`, implying the default behavior which is `yes`.
- **Recommendation:** None

Rule 6.1.2.1: Ensure rsyslog is installed

- **Status:** PASS
- **Evidence:** `rsyslog_installed` is `true`.
- **Recommendation:** None

Rule 6.1.2.2: Ensure rsyslog service is enabled and active

- **Status:** PASS
- **Evidence:** `rsyslog` service is enabled and active.
- **Recommendation:** None

Rule 6.1.2.3: Ensure rsyslog default file permissions are configured

- **Status:** PASS
- **Evidence:** `/etc/rsyslog.conf` sets `$FileCreateMode 0640`.
- **Recommendation:** None

Rule 6.1.2.4: Ensure rsyslog logging is configured

- **Status:** INFORMATIONAL
- **Evidence:** Rsyslog is configured with default rules in `/etc/rsyslog.d/50-default.conf` and additional configurations in `20-ufw.conf` and `21-cloudinit.conf`.
- **Recommendation:** Manually review the rsyslog configuration files to ensure all necessary logs are being captured and processed according to organizational policy.

Rule 6.1.2.5: Ensure rsyslog is configured to send logs to a remote log host

- **Status:** INFORMATIONAL
- **Evidence:** Rsyslog is not explicitly configured to send logs to a remote log host; relevant lines are commented out in `/etc/rsyslog.conf`.
- **Recommendation:** If remote logging is required, configure rsyslog to send logs to a central log server by uncommenting and configuring remote forwarding rules in `/etc/rsyslog.conf` or a file in `/etc/rsyslog.d/`.

Rule 6.1.2.6: Ensure rsyslog is not configured to receive logs from a remote client

- **Status:** PASS
- **Evidence:** `imudp` and `imtcp` modules for receiving remote logs are commented out in `/etc/rsyslog.conf`.
- **Recommendation:** None

Rule 6.1.2.7: Ensure logrotate is configured

- **Status:** INFORMATIONAL
- **Evidence:** `logrotate` is installed, its timer is enabled and active, and various log rotation configurations are present in `/etc/logrotate.conf` and `/etc/logrotate.d/`.
- **Recommendation:** Manually review the `logrotate` configurations to ensure they meet organizational requirements for log retention and disk usage.

Rule 6.1.2.8: Ensure rsyslog-gnutls is installed

- **Status:** FAIL
- **Evidence:** `rsyslog_gnutls_installed` is false.
- **Recommendation:** Install the `rsyslog-gnutls` package using `sudo apt install rsyslog-gnutls` to enable secure remote log forwarding.

Rule 6.1.2.9: Ensure rsyslog is configured to use gtls for forwarding

- **Status:** FAIL
- **Evidence:** `rsyslog-gnutls` is not installed, so GTLS forwarding cannot be configured.
- **Recommendation:** Install `rsyslog-gnutls` and then configure rsyslog to use GTLS for secure remote log forwarding.

Rule 6.1.2.10: Ensure rsyslog CA certificates are configured

- **Status:** INFORMATIONAL
- **Evidence:** `rsyslog-gnutls` is not installed, so CA certificates for GTLS forwarding are not applicable.
- **Recommendation:** Install `rsyslog-gnutls` and then configure CA certificates for secure remote log forwarding.

Rule 6.1.3.1: Ensure access to all logfiles has been configured

- **Status:** INFORMATIONAL
- **Evidence:** A list of log files and their permissions is provided. Examples include `/var/log/auth.log` (0640 root adm), `/var/log/syslog` (0640 syslog adm), `/var/log/`

wtmp (0664 root utmp).

- **Recommendation:** Manually review the permissions of all log files to ensure they are appropriately restricted to prevent unauthorized access to sensitive log data.

SYSTEM AUDITING (auditd) Rules

Rule 6.2.1.1.1: Ensure auditd is installed

- **Status:** FAIL
- **Evidence:** `auditd_installed` is false.
- **Recommendation:** Install the auditd package using `sudo apt install auditd`.

Rule 6.2.1.1.2: Ensure auditd service is enabled and active

- **Status:** FAIL
- **Evidence:** `auditd` is not installed, so its service cannot be enabled or active.
- **Recommendation:** Install `auditd` and then enable and start the `auditd` service.

Rule 6.2.1.1.3: Ensure auditing for processes that start prior to auditd is enabled

- **Status:** FAIL
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and configure GRUB to pass `audit=1` to the kernel.

Rule 6.2.1.1.4: Ensure audit_backlog_limit is sufficient

- **Status:** FAIL
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and configure `audit_backlog_limit` in `/etc/audit/audit.conf` to an appropriate value (e.g., 8192).

Rule 6.2.1.2.1: Ensure audit log storage size is configured

- **Status:** FAIL
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and configure `max_log_file` in `/etc/audit/audit.conf` to an appropriate size (e.g., 50MB).

Rule 6.2.1.2.2: Ensure audit logs are not automatically deleted

- **Status:** FAIL
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and set `max_log_file_action = keep_logs` in `/etc/audit/audit.conf`.

Rule 6.2.1.2.3: Ensure system is disabled when audit logs are full

- **Status:** FAIL
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and set `space_left_action = SYSLOG` and

```
admin_space_left_action = halt in /etc/audit/audit.conf.
```

Rule 6.2.1.2.4: Ensure system warns when audit logs are low on space

- **Status:** FAIL
- **Evidence:** `auditd` is not installed.
- **Recommendation:** Install `auditd` and set `space_left = 75` and `action_mail_acct = root` in `/etc/audit/audit.conf`.

Rule 6.2.3.1-6.2.3.30: Ensure auditd rules cover sudoers, identity, logins, session, time-change, MAC policy, access, mounts, deletion, loading modules, and configuration is immutable

- **Status:** FAIL
- **Evidence:** `auditd` is not installed, so no audit rules are in place.
- **Recommendation:** Install `auditd` and configure the comprehensive set of audit rules as specified in the CIS Benchmark (Section 6.2.3).

Rule 6.2.4.1-6.2.4.10: Ensure audit log files and audit tools file access and permissions are strictly constrained

- **Status:** FAIL
- **Evidence:** `auditd` is not installed, so its log files and tools are not present or secured as per `auditd` requirements.
- **Recommendation:** Install `auditd` and ensure that audit log files (e.g., `/var/log/audit/audit.log`) and audit tools (e.g., `/sbin/auditctl`, `/sbin/audispd`) have appropriate restrictive permissions (e.g., 0640 for logs, 0755 for tools, owned by `root:root`).

INTEGRITY CHECKING Rules

Rule 6.3.1: Ensure AIDE is installed

- **Status:** FAIL
- **Evidence:** `aide_installed` is false.
- **Recommendation:** Install the `aide` package using `sudo apt install aide aide-common`.

Rule 6.3.2: Ensure filesystem integrity is regularly checked

- **Status:** FAIL
- **Evidence:** `AIDE` is not installed, so no integrity checks are scheduled.
- **Recommendation:** Install `AIDE` and configure a cron job or systemd timer to regularly run `aide --check`.

Rule 6.3.3: Ensure cryptographic mechanisms protect tools

- **Status:** FAIL
- **Evidence:** `AIDE` is not installed, so its tools are not protected by `AIDE` itself.
- **Recommendation:** Install `AIDE` and configure its database to include checks for `AIDE` binaries and configuration files using strong hashing algorithms (e.g., `sha512`).

LOCAL USER Rules

Rule 7.2.1: Ensure accounts in `/etc/passwd` use shadowed passwords

- **Status:** PASS
- **Evidence:** `/etc/shadow` exists with restricted permissions (0640), and `pam_unix.so` is configured for password hashing.
- **Recommendation:** None

Rule 7.2.2: Ensure `/etc/shadow` password fields are not empty

- **Status:** PASS
- **Evidence:** `root_shadow_password_status` is `*`, and other user entries in `user_chage_info` imply non-empty password fields.
- **Recommendation:** None

Rule 7.2.3: Ensure all groups in `/etc/passwd` exist in `/etc/group`

- **Status:** PASS
- **Evidence:** `orphan_gids_in_passwd` is empty.
- **Recommendation:** None

Rule 7.2.4: Ensure shadow group is empty

- **Status:** PASS
- **Evidence:** `shadow_group_members` is empty.
- **Recommendation:** None

Rule 7.2.5: Ensure no duplicate UIDs exist

- **Status:** PASS
- **Evidence:** `duplicate_uids` is an empty object.
- **Recommendation:** None

Rule 7.2.6: Ensure no duplicate GIDs exist

- **Status:** PASS
- **Evidence:** `duplicate_gids` is an empty object.
- **Recommendation:** None

Rule 7.2.7: Ensure no duplicate user names exist

- **Status:** PASS
- **Evidence:** `duplicate_usernames` is an empty object.
- **Recommendation:** None

Rule 7.2.8: Ensure no duplicate group names exist

- **Status:** PASS
- **Evidence:** `duplicate_groupnames` is an empty object.
- **Recommendation:** None

Rule 7.2.9: Ensure local interactive user home directories are configured

- **Status:** PASS
- **Evidence:** `home_directory_issues` is empty.
- **Recommendation:** None

Rule 7.2.10: Ensure local interactive user dot files access is configured

- **Status:** PASS
- **Evidence:** `dotfile_permission_issues` is empty.
- **Recommendation:** None

SECTION 1.1 – FILESYSTEM KERNEL MODULES & PARTITIONS

Rule 1.1.1.1: Ensure cramfs filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `cramfs` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.2: Ensure freevxfs filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `freevxfs` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.3: Ensure hfs filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `hfs` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.4: Ensure hfsplus filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `hfsplus` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.5: Ensure jffs2 filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error,

preventing evaluation.

- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `jffs2` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.6: Ensure overlay filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `overlay` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.7: Ensure squashfs filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `squashfs` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.8: Ensure udf filesystem module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `udf` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.9: Ensure firewire-core module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `firewire-core` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.10: Ensure usb-storage module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `usb-storage` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 1.1.1.11: Unused filesystems

- **Status:** INFORMATIONAL
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing the listing of loaded kernel modules.
- **Recommendation:** Manually review the output of `lsmod` to identify any unused

filesystem modules that should be disabled.

Rule 1.1.2.1.1: Ensure /tmp is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure `/tmp` is configured as a separate partition or a `tmpfs` filesystem.

Rule 1.1.2.1.2: Ensure nodev option is set on /tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nodev` mount option is set for `/tmp`.

Rule 1.1.2.1.3: Ensure nosuid option is set on /tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nosuid` mount option is set for `/tmp`.

Rule 1.1.2.1.4: Ensure noexec option is set on /tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `noexec` mount option is set for `/tmp`.

Rule 1.1.2.2.1: Ensure /dev/shm is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure `/dev/shm` is configured as a separate partition or a `tmpfs` filesystem.

Rule 1.1.2.2.2: Ensure nodev option is set on /dev/shm partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nodev` mount option is set for `/dev/shm`.

Rule 1.1.2.2.3: Ensure nosuid option is set on /dev/shm partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nosuid` mount option is set for `/dev/shm`.

Rule 1.1.2.2.4: Ensure noexec option is set on /dev/shm partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `noexec` mount option is set for `/dev/shm`.

Rule 1.1.2.3.1: Ensure /home is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure `/home` is configured as a separate partition.

Rule 1.1.2.3.2: Ensure nodev option is set on /home partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nodev` mount option is set for `/home`.

Rule 1.1.2.3.3: Ensure nosuid option is set on /home partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nosuid` mount option is set for `/home`.

Rule 1.1.2.4.1: Ensure /var is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure `/var` is configured as a separate partition.

Rule 1.1.2.4.2: Ensure nodev option is set on /var partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nodev`

mount option is set for `/var`.

Rule 1.1.2.4.3: Ensure nosuid option is set on /var partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nosuid` mount option is set for `/var`.

Rule 1.1.2.5.1: Ensure /var/tmp is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure `/var/tmp` is configured as a separate partition.

Rule 1.1.2.5.2: Ensure nodev option is set on /var/tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nodev` mount option is set for `/var/tmp`.

Rule 1.1.2.5.3: Ensure nosuid option is set on /var/tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nosuid` mount option is set for `/var/tmp`.

Rule 1.1.2.5.4: Ensure noexec option is set on /var/tmp partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `noexec` mount option is set for `/var/tmp`.

Rule 1.1.2.6.1: Ensure /var/log is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure `/var/log` is configured as a separate partition.

Rule 1.1.2.6.2: Ensure nodev option is set on /var/log partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nodev` mount option is set for `/var/log`.

Rule 1.1.2.6.3: Ensure nosuid option is set on /var/log partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nosuid` mount option is set for `/var/log`.

Rule 1.1.2.6.4: Ensure noexec option is set on /var/log partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `noexec` mount option is set for `/var/log`.

Rule 1.1.2.7.1: Ensure /var/log/audit is a separate partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure `/var/log/audit` is configured as a separate partition.

Rule 1.1.2.7.2: Ensure nodev option is set on /var/log/audit partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nodev` mount option is set for `/var/log/audit`.

Rule 1.1.2.7.3: Ensure nosuid option is set on /var/log/audit partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `nosuid` mount option is set for `/var/log/audit`.

Rule 1.1.2.7.4: Ensure noexec option is set on /var/log/audit partition

- **Status:** UNKNOWN
- **Evidence:** The `filesystem` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `filesystem` data collection. Ensure the `noexec`

mount option is set for `/var/log/audit`.

SECTION 1.2 – PACKAGE MANAGEMENT

Rule 1.2.1.1: Signed-By option

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Manually review `APT.sources.list` and `sources.list.d` files to ensure all repositories use the `Signed-By` option.

Rule 1.2.1.2: Weak dependencies

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Configure APT to not install recommended or suggested packages by adding `APT::Install-Recommends "false";` and `APT::Install-Suggests "false";` to `/etc/apt/apt.conf.d/`.

Rule 1.2.1.3: GPG key files access

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Ensure GPG key files in `/etc/apt/trusted.gpg.d/` and `/usr/share/keyrings/` have permissions `0644` or less, owned by `root:root`.

Rule 1.2.1.4: `/etc/apt/trusted.gpg.d` dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Ensure `/etc/apt/trusted.gpg.d` has permissions `0755` or less, owned by `root:root`.

Rule 1.2.1.5: `/etc/apt/auth.conf.d` dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Ensure `/etc/apt/auth.conf.d` either does not exist or has permissions `0700` or less, owned by `root:root`.

Rule 1.2.1.6: files in `auth.conf.d`

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Ensure all files in `/etc/apt/auth.conf.d` have permissions `0600` or less, owned by `root:root`.

Rule 1.2.1.7: `/usr/share/keyrings` dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Ensure `/usr/share/keyrings` has permissions `0755` or less, owned by `root:root`.

Rule 1.2.1.8: `/etc/apt/sources.list.d` dir

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Ensure `/etc/apt/sources.list.d` has permissions `0755` or less, owned by `root:root`.

Rule 1.2.1.9: files in `sources.list.d`

- **Status:** UNKNOWN
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Ensure all files in `/etc/apt/sources.list.d` have permissions `0644` or less, owned by `root:root`.

Rule 1.2.2.1: Package updates installed

- **Status:** INFORMATIONAL
- **Evidence:** The `package_management` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `package_management` data collection. Manually check for available package updates using `sudo apt update && sudo apt list --upgradable` and ensure unattended upgrades are configured and working.

SECTION 1.3 – APPARMOR

Rule 1.3.1.1: AppArmor packages installed

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `apparmor` data collection. Ensure `apparmor` and `apparmor-utils` packages are installed.

Rule 1.3.1.2: AppArmor enabled

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `apparmor` data collection. Ensure AppArmor is enabled in the kernel (via GRUB) and its service is enabled and active.

Rule 1.3.1.3: All AppArmor profiles enforcing

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `apparmor` data collection. Ensure all AppArmor profiles are in enforcing mode and no unconfined processes are running.

Rule 1.3.1.4: `apparmor_restrict_unprivileged_unconfined`

- **Status:** UNKNOWN
- **Evidence:** The `apparmor` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `apparmor` data collection. Ensure `apparmor_restrict_unprivileged_unconfined` is set to 1 at runtime and persisted.

SECTION 1.4 – BOOTLOADER

Rule 1.4.1: Bootloader password

- **Status:** UNKNOWN
- **Evidence:** The `bootloader` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `bootloader` data collection. Configure a password for the GRUB bootloader to prevent unauthorized modification of boot parameters.

Rule 1.4.2: Bootloader permissions

- **Status:** UNKNOWN
- **Evidence:** The `bootloader` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `bootloader` data collection. Ensure GRUB configuration files (e.g., `/boot/grub/grub.cfg`) have permissions 0400 or 0600, owned by `root:root`.

SECTION 1.5 – PROCESS HARDENING

Rule 1.5.1: `fs.protected_hardlinks`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.

- **Recommendation:** Investigate the `process_hardening` data collection. Set `fs.protected_hardlinks = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 1.5.2: `fs.protected_symlinks`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Set `fs.protected_symlinks = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 1.5.3: `kernel.yama.ptrace_scope`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Set `kernel.yama.ptrace_scope = 1` (or higher) at runtime and persist it in `/etc/sysctl.d/`.

Rule 1.5.4: `fs.suid_dumpable`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Set `fs.suid_dumpable = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 1.5.5: `kernel.dmesg_restrict`

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Set `kernel.dmesg_restrict = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 1.5.6: Prelink not installed

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Ensure the `prelink` package is not installed.

Rule 1.5.7: Appport disabled

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Ensure Appport is not installed or is disabled.

Rule 1.5.8: kernel.kptr_restrict

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Set `kernel.kptr_restrict = 2` (or 1) at runtime and persist it in `/etc/sysctl.d/`.

Rule 1.5.9: kernel.randomize_va_space

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Set `kernel.randomize_va_space = 2` at runtime and persist it in `/etc/sysctl.d/`.

Rule 1.5.10: Core dumps restricted

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Set `ulimit -H -c 0` in `/etc/security/limits.conf` and ensure hard core 0 is present.

Rule 1.5.11: systemd-coredump ProcessSizeMax

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Configure `ProcessSizeMax=0` in `/etc/systemd/coredump.conf` to disable core dumps.

Rule 1.5.12: systemd-coredump Storage

- **Status:** UNKNOWN
- **Evidence:** The `process_hardening` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `process_hardening` data collection. Configure `Storage=none` in `/etc/systemd/coredump.conf` to disable core dump storage.

SECTION 1.6 – WARNING BANNERS

Rule 1.6.1: /etc/motd configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Configure `/etc/motd` with a compliant warning banner that does not contain OS/kernel release information.

Rule 1.6.2: /etc/issue configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Configure `/etc/issue` with a compliant warning banner that does not contain OS/kernel release information.

Rule 1.6.3: /etc/issue.net configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Configure `/etc/issue.net` with a compliant warning banner that does not contain OS/kernel release information.

Rule 1.6.4: pam_motd configured

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Ensure `pam_motd` is configured to display a warning banner and dynamic motd scripts are reviewed or disabled.

Rule 1.6.6: access to /etc/motd

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Ensure `/etc/motd` has permissions `0644` or less, owned by `root:root`.

Rule 1.6.7: access to /etc/issue

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Ensure `/etc/issue` has permissions `0644` or less, owned by `root:root`.

Rule 1.6.8: access to /etc/issue.net

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Ensure `/etc/issue.net` has permissions `0644` or less, owned by `root:root`.

Rule 1.6.9: access to pam_motd files

- **Status:** UNKNOWN
- **Evidence:** The `warning_banners` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `warning_banners` data collection. Ensure `pam_motd` related files (e.g., `/etc/update-motd.d/*`) have permissions `0644` or `0755` (for scripts), owned by `root:root`.

SECTION 1.7 – GNOME DISPLAY MANAGER

Rule 1.7.1: GDM login banner

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `gnome` data collection. If GDM is installed, configure a login banner by setting `banner-message-enable=true` and `banner-message-text` in GDM's dconf settings.

Rule 1.7.2: disable-user-list

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `gnome` data collection. If GDM is installed, set `disable-user-list=true` in GDM's dconf settings to prevent listing users on the login screen.

Rule 1.7.3: GDM screen lock

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `gnome` data collection. If GDM is installed, ensure screen lock is enabled (`lock-enabled=true`) and the lock delay is configured (e.g., `lock-delay <= 900` seconds), with dconf locks in place.

Rule 1.7.4: GDM automount

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `gnome` data collection. If GDM is installed, disable automounting by setting `automount=false` and `automount-open=false` in GDM's dconf settings, with dconf locks in place.

Rule 1.7.5: GDM autorun-never

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `gnome` data collection. If GDM is installed, set

`autorun-never=true` in GDM's dconf settings, with a dconf lock in place.

Rule 1.7.6: XDMCP disabled

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `gnome` data collection. If GDM is installed, ensure XDMCP is disabled by verifying `Enable=true` is not present under `[xdmcp]` in `gdm3-custom.conf`.

Rule 1.7.7: Xwayland configured

- **Status:** UNKNOWN
- **Evidence:** The `gnome` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `gnome` data collection. If GDM is installed, ensure Xwayland is properly configured (e.g., `WaylandEnable=false` or `DisableXwayland=true`).

SECTION 2 – SERVICES

Rule 2.1.1: Ensure autofs is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `autofs` package is not installed and its service is not active.

Rule 2.1.2: Mail Transfer Agent local-only mode

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. If an MTA is installed, ensure it is configured to listen only on the loopback interface (e.g., `inet_interfaces = loopback-only` for Postfix).

Rule 2.1.3: Ensure avahi server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `avahi-daemon` package is not installed and its service is not active.

Rule 2.1.4: Only approved services listening

- **Status:** INFORMATIONAL
- **Evidence:** The `services` object in the security profile has a `json_parse` error,

preventing the listing of listening sockets.

- **Recommendation:** Manually review listening network services using `ss -tln` or `netstat -tln` to ensure only approved services are listening.

Rule 2.1.5: Ensure dhcp server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `isc-dhcp-server` package is not installed and its service is not active.

Rule 2.1.6: Ensure web server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure no web server packages (e.g., `apache2`, `nginx`) are installed and no web server units are active.

Rule 2.1.7: Ensure dns/bind9 server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `bind9` package is not installed and its service is not active.

Rule 2.1.9: Ensure dnsmasq is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `dnsmasq` package is not installed and its service is not active.

Rule 2.1.10: Ensure ldap/slapd server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `slapd` package is not installed and its service is not active.

Rule 2.1.11: Ensure imap/pop3 server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure no IMAP/POP3 server packages (e.g., `dovecot`, `cyrus-imapd`) are installed and no IMAP/POP3 server units are active.

Rule 2.1.12: Ensure nfs server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `nfs-kernel-server` package is not installed and its service is not active.

Rule 2.1.13: Ensure nis/ypserv server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `nis` or `ypserv` packages are not installed and their services are not active.

Rule 2.1.14: Ensure cups print server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `cups` package is not installed and its service is not active.

Rule 2.1.15: Ensure rpcbind is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `rpcbind` package is not installed and its service is not active.

Rule 2.1.16: Ensure rsync daemon is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `rsync` package is not installed and its service is not active.

Rule 2.1.17: Ensure samba/smbd is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `samba` package is not installed and its service is not active.

Rule 2.1.18: Ensure snmpd is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error,

preventing evaluation.

- **Recommendation:** Investigate the `services` data collection. Ensure `snmpd` package is not installed and its service is not active.

Rule 2.1.21: Ensure squid proxy is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `squid` package is not installed and its service is not active.

Rule 2.1.22: Ensure xinetd is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `xinetd` package is not installed and its service is not active.

Rule 2.1.23: Ensure X window server is not in use

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure X Window System packages are not installed and no X server processes are running.

Rule 2.2.1: Ensure nis/ypbind client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `nis` or `ypbind` client packages are not installed.

Rule 2.2.2: Ensure rsh-client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `rsh-client` package is not installed.

Rule 2.2.3: Ensure talk client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `talk` client package is not installed.

Rule 2.2.5: Ensure ldap-utils client is not installed

- **Status:** UNKNOWN
- **Evidence:** The `services` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `services` data collection. Ensure `ldap-utils` package is not installed.

Rule 2.3.1.1: Single time sync daemon in use

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `time_sync` data collection. Ensure only one time synchronization daemon (e.g., `systemd-timesyncd` or `chrony`) is active.

Rule 2.3.2.1: systemd-timesyncd timeserver configured

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `time_sync` data collection. If `systemd-timesyncd` is used, ensure `NTP=` or `FallbackNTP=` is configured in `/etc/systemd/timesyncd.conf`.

Rule 2.3.2.2: systemd-timesyncd enabled and running

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `time_sync` data collection. If `systemd-timesyncd` is used, ensure its service is enabled and active.

Rule 2.3.3.1: chrony timeserver configured

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `time_sync` data collection. If `chrony` is used, ensure `server` or `pool` directives are configured in `/etc/chrony/chrony.conf`.

Rule 2.3.3.2: chrony process user

- **Status:** UNKNOWN
- **Evidence:** The `time_sync` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `time_sync` data collection. If `chrony` is used, ensure it runs as the `_chrony` user.

Rule 2.3.3.3: chrony enabled and running

- **Status:** UNKNOWN

- **Evidence:** The `time_sync` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `time_sync` data collection. If `chrony` is used, ensure its service is enabled and active.

Rule 2.4.1.1: cron daemon enabled and active

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `cron` package is installed and its service is enabled and active.

Rule 2.4.1.2: access to /etc/crontab

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/etc/crontab` has permissions `0600` or `0640`, owned by `root:root`.

Rule 2.4.1.3: access to /etc/cron.hourly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/etc/cron.hourly` has permissions `0700` or `0750`, owned by `root:root`.

Rule 2.4.1.4: access to /etc/cron.daily

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/etc/cron.daily` has permissions `0700` or `0750`, owned by `root:root`.

Rule 2.4.1.5: access to /etc/cron.weekly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/etc/cron.weekly` has permissions `0700` or `0750`, owned by `root:root`.

Rule 2.4.1.6: access to /etc/cron.monthly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/etc/cron.monthly` has permissions `0700` or `0750`, owned by `root:root`.

Rule 2.4.1.7: access to /etc/cron.yearly

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/etc/cron.yearly` has permissions `0700` or `0750`, owned by `root:root`.

Rule 2.4.1.8: access to /etc/cron.d

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/etc/cron.d` has permissions `0700` or `0750`, owned by `root:root`.

Rule 2.4.1.9: access to crontab mechanism

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `/var/spool/cron/crontabs` has permissions `1730` or `0700/0755`, owned by `root:crontab` or `root:root`, and `crontab` binary is owned by `root`.

Rule 2.4.2.1: access to at configured

- **Status:** UNKNOWN
- **Evidence:** The `job_schedulers` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `job_schedulers` data collection. Ensure `at.allow` exists with owner `root` (or `daemon`), group `root` (or `daemon`), mode `<= 0640`, and `at.deny` does not exist or is empty.

SECTION 3 – NETWORK CONFIGURATION

Rule 3.1.1: IPv6 status identified

- **Status:** INFORMATIONAL
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Manually identify the IPv6 status (enabled/disabled) on the system.

Rule 3.1.2: Ensure wireless interfaces are not available

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Ensure no wireless interfaces are detected and no wireless kernel modules are loaded.

Rule 3.1.3: Ensure Bluetooth services are not in use

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Ensure `bluetooth` package is not installed or its service is disabled and inactive.

Rule 3.2.1: Ensure atm kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Ensure the `atm` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 3.2.2: Ensure can kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Ensure the `can` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 3.2.3: Ensure dccp kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Ensure the `dccp` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 3.2.4: Ensure rds kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Ensure the `rds` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 3.2.5: Ensure sctp kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Ensure the `sctp` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 3.2.6: Ensure tipc kernel module is disabled

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse`

error, preventing evaluation.

- **Recommendation:** Investigate the `network_config` data collection. Ensure the `tipc` kernel module is not loaded and is blacklisted/disabled via `modprobe` configuration.

Rule 3.3.1.1: Ensure `net.ipv4.ip_forward` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.ip_forward = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.2: Ensure `net.ipv4.conf.all.forwarding` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.all.forwarding = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.3: Ensure `net.ipv4.conf.default.forwarding` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.default.forwarding = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.4: Ensure `net.ipv4.conf.all.send_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.all.send_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.5: Ensure `net.ipv4.conf.default.send_redirects` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.default.send_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.6: Ensure `net.ipv4.icmp_ignore_bogus_error_responses` is configured

- **Status:** UNKNOWN

- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.icmp_ignore_bogus_error_responses = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.7: Ensure net.ipv4.icmp_echo_ignore_broadcasts is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.icmp_echo_ignore_broadcasts = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.8: Ensure net.ipv4.conf.all.accept_redirects is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.all.accept_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.9: Ensure net.ipv4.conf.default.accept_redirects is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.default.accept_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.10: Ensure net.ipv4.conf.all.secure_redirects is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.all.secure_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.11: Ensure net.ipv4.conf.default.secure_redirects is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.default.secure_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.12: Ensure net.ipv4.conf.all.rp_filter is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.all.rp_filter = 1` (or 2) at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.13: Ensure `net.ipv4.conf.default.rp_filter` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.default.rp_filter = 1` (or 2) at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.14: Ensure `net.ipv4.conf.all.accept_source_route` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.all.accept_source_route = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.15: Ensure `net.ipv4.conf.default.accept_source_route` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.default.accept_source_route = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.16: Ensure `net.ipv4.conf.all.log_martians` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.all.log_martians = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.17: Ensure `net.ipv4.conf.default.log_martians` is configured

- **Status:** UNKNOWN
- **Evidence:** The `network_config` object in the security profile has a `json_parse` error, preventing evaluation.
- **Recommendation:** Investigate the `network_config` data collection. Set `net.ipv4.conf.default.log_martians = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.1.18: Ensure net.ipv4.tcp_syncookies is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv4.tcp_syncookies = 1` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.1: Ensure net.ipv6.conf.all.forwarding is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.all.forwarding = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.2: Ensure net.ipv6.conf.default.forwarding is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.default.forwarding = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.3: Ensure net.ipv6.conf.all.accept_redirects is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.all.accept_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.4: Ensure net.ipv6.conf.default.accept_redirects is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.default.accept_redirects = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.5: Ensure net.ipv6.conf.all.accept_source_route is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.all.accept_source_route = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.6: Ensure net.ipv6.conf.default.accept_source_route is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.default.accept_source_route = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.7: Ensure net.ipv6.conf.all.accept_ra is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.all.accept_ra = 0` at runtime and persist it in `/etc/sysctl.d/`.

Rule 3.3.2.8: Ensure net.ipv6.conf.default.accept_ra is configured

- **Status:** UNKNOWN
- **Evidence:** The network_config object in the security profile has a json_parse error, preventing evaluation.
- **Recommendation:** Investigate the network_config data collection. Set `net.ipv6.conf.default.accept_ra = 0` at runtime and persist it in `/etc/sysctl.d/`.

SECTION 5.3 – PLUGGABLE AUTHENTICATION MODULES (PAM)

Rule 5.3.1.1: libpam-runtime: installed and up-to-date

- **Status:** PASS
- **Evidence:** libpam-runtime is installed (true) and installed_version (1.5.3-5ubuntu5.5) matches candidate_version (1.5.3-5ubuntu5.5).
- **Recommendation:** None

Rule 5.3.1.2: libpam-modules: installed and up-to-date

- **Status:** PASS
- **Evidence:** libpam-modules is installed (true) and installed_version (1.5.3-5ubuntu5.5) matches candidate_version (1.5.3-5ubuntu5.5).
- **Recommendation:** None

Rule 5.3.1.3: libpam-pwquality: installed and up-to-date

- **Status:** FAIL
- **Evidence:** libpam-pwquality is not installed (installed: false).
- **Recommendation:** Install the libpam-pwquality package using `sudo apt install libpam-pwquality`.

Rule 5.3.1.4: cracklib-runtime: installed and up-to-date

- **Status:** FAIL
- **Evidence:** cracklib-runtime is not installed (installed: false).
- **Recommendation:** Install the cracklib-runtime package using `sudo apt install cracklib-runtime`.

Rule 5.3.2.1: pam_unix enabled

- **Status:** PASS
- **Evidence:** pam_unix.so module lines are present in common-auth, common-account, common-password, and common-session.
- **Recommendation:** None

Rule 5.3.2.2: pam_faillock enabled

- **Status:** FAIL
- **Evidence:** No pam_faillock.so module lines are present in the PAM configuration.
- **Recommendation:** Configure pam_faillock.so in /etc/pam.d/common-auth and /etc/pam.d/common-account to lock accounts after failed login attempts.

Rule 5.3.2.3: pam_pwquality enabled

- **Status:** FAIL
- **Evidence:** No pam_pwquality.so module lines are present in the PAM configuration.
- **Recommendation:** Configure pam_pwquality.so in /etc/pam.d/common-password to enforce password quality requirements.

Rule 5.3.2.4: pam_pwhistory enabled

- **Status:** FAIL
- **Evidence:** No pam_pwhistory.so module lines are present in the PAM configuration.
- **Recommendation:** Configure pam_pwhistory.so in /etc/pam.d/common-password to enforce password history.

Rule 5.3.3.1.1: pam_faillock deny <= 5

- **Status:** FAIL
- **Evidence:** pam_faillock is not enabled, so deny is not configured.
- **Recommendation:** Enable pam_faillock.so and set deny = 5 or less in /etc/security/faillock.conf or inline PAM configuration.

Rule 5.3.3.1.2: pam_faillock unlock_time >= 900

- **Status:** FAIL
- **Evidence:** pam_faillock is not enabled, so unlock_time is not configured.
- **Recommendation:** Enable pam_faillock.so and set unlock_time = 900 or unlock_time = 0 in /etc/security/faillock.conf or inline PAM configuration.

Rule 5.3.3.1.3: pam_faillock even_deny_root / root_unlock_time

- **Status:** FAIL
- **Evidence:** `pam_faillock` is not enabled, so `even_deny_root` and `root_unlock_time` are not configured.
- **Recommendation:** Enable `pam_faillock.so` and add `even_deny_root` and `root_unlock_time = 60` (or higher) in `/etc/security/faillock.conf` or inline PAM configuration.

Rule 5.3.3.2.1: pam_pwquality difok >= 2

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, so `difok` is not configured.
- **Recommendation:** Enable `pam_pwquality.so` and set `difok = 2` or more in `/etc/security/pwquality.conf` or inline PAM configuration.

Rule 5.3.3.2.2: pam_pwquality minlen >= 14

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, so `minlen` is not configured.
- **Recommendation:** Enable `pam_pwquality.so` and set `minlen = 14` or more in `/etc/security/pwquality.conf` or inline PAM configuration.

Rule 5.3.3.2.3: Password complexity (dcredit, ucredit, lcredit, ocredit)

- **Status:** INFORMATIONAL
- **Evidence:** `dcredit`, `ucredit`, `lcredit`, `ocredit` are not explicitly configured in `pwquality.conf` or inline PAM.
- **Recommendation:** Manually review and consider setting `dcredit`, `ucredit`, `lcredit`, `ocredit` to enforce password complexity requirements (e.g., -1 for at least one digit, uppercase, lowercase, other character).

Rule 5.3.3.2.4: pam_pwquality maxrepeat <= 3

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, so `maxrepeat` is not configured.
- **Recommendation:** Enable `pam_pwquality.so` and set `maxrepeat = 3` or less in `/etc/security/pwquality.conf` or inline PAM configuration.

Rule 5.3.3.2.5: pam_pwquality maxsequence <= 3

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, so `maxsequence` is not configured.
- **Recommendation:** Enable `pam_pwquality.so` and set `maxsequence = 3` or less in `/etc/security/pwquality.conf` or inline PAM configuration.

Rule 5.3.3.2.6: pam_pwquality dictcheck = 1

- **Status:** FAIL
- **Evidence:** `pam_pwquality` is not enabled, so `dictcheck` is not configured.
- **Recommendation:** Enable `pam_pwquality.so` and set `dictcheck = 1` in `/etc/security/pwquality.conf` or inline PAM configuration.

Rule 5.3.3.2.7: pam_pwquality enforcing = 1

- **Status:** FAIL
- **Evidence:** pam_pwquality is not enabled, so enforcing is not configured.
- **Recommendation:** Enable pam_pwquality.so and set enforcing = 1 in /etc/security/pwquality.conf or inline PAM configuration.

Rule 5.3.3.2.8: pam_pwquality enforce_for_root present

- **Status:** FAIL
- **Evidence:** pam_pwquality is not enabled, so enforce_for_root is not configured.
- **Recommendation:** Enable pam_pwquality.so and add enforce_for_root in /etc/security/pwquality.conf or inline PAM configuration.

Rule 5.3.3.3.1: pam_pwhistory remember >= 24

- **Status:** FAIL
- **Evidence:** pam_pwhistory is not enabled, so remember is not configured.
- **Recommendation:** Enable pam_pwhistory.so and set remember = 24 or more in /etc/pam.d/common-password.

Rule 5.3.3.3.2: pam_pwhistory enforce_for_root present

- **Status:** FAIL
- **Evidence:** pam_pwhistory is not enabled, so enforce_for_root is not configured.
- **Recommendation:** Enable pam_pwhistory.so and add enforce_for_root in /etc/pam.d/common-password.

Rule 5.3.3.3.3: pam_pwhistory use_authok present

- **Status:** FAIL
- **Evidence:** pam_pwhistory is not enabled, so use_authok is not configured.
- **Recommendation:** Enable pam_pwhistory.so and add use_authok in /etc/pam.d/common-password.

Rule 5.3.3.4.1: pam_unix nullok absent

- **Status:** FAIL
- **Evidence:** nullok is present in pam_unix.so line in common-auth.
- **Recommendation:** Remove nullok from the pam_unix.so line in /etc/pam.d/common-auth to prevent authentication with empty passwords.

Rule 5.3.3.4.2: pam_unix remember absent

- **Status:** PASS
- **Evidence:** remember is not present in any pam_unix.so line.
- **Recommendation:** None

Rule 5.3.3.4.3: pam_unix Strong hashing

- **Status:** PASS
- **Evidence:** yescrypt is present in the pam_unix.so line in common-password.
- **Recommendation:** None

Rule 5.3.3.4.4: pam_unix use_authok present (in password stack)

- **Status:** FAIL
 - **Evidence:** use_authok is not present in the pam_unix.so line in common-password.
 - **Recommendation:** Add use_authok to the pam_unix.so line in /etc/pam.d/common-password.
-

SUID/SGID Triage Report

```
{
  "findings": [],
  "summary": {
    "suspicious_found": 0,
    "message": "No unexpected SUID/SGID files detected."
  }
}
```